

Sigvard Bratlie

Data Scientist & AI Engineer

Building intelligent systems — from legal agents to market valuation platforms

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About Me

I'm passionate about **LLMs**, **computer vision**, **data pipelines** and turning complex data into actionable insights or autonomous agents.

Currently exploring agentic AI, multimodal models, scalable ETL and production-grade ML systems — mostly in Python, with heavy use of Google Cloud, LangChain/LangGraph and modern MLOps.

Master's thesis focused on **Agentic Legal AI** — building LLM-powered assistants for lawyers.



Project Portfolio

Here are some of my key projects:



Agentic Legal AI (Master's Thesis)

LLM-powered legal case management agent

Intelligent agent that helps lawyers with case summarization, structured fact extraction, document analysis, RAG on Norwegian law and real-time chat assistance.

- **Tech stack:** LangGraph, LangChain, Gemini / GPT / Claude, FastAPI, Streamlit, Supabase (Auth + PostgreSQL + Storage), BigQuery Vector Store, Tavily Search
- **Highlights:** Structured Factsheet (Pydantic), multi-phase document processing pipeline, streaming SSE chat, Supabase auth + JWT
- **GitHub:** [sigvardbratlie/master-thesis](https://github.com/sigvardbratlie/master-thesis)



Weather & Electricity Dashboard

Interactive multi-page Streamlit dashboard exploring Norwegian electricity production/consumption vs historical weather data.

- **Features:** Time-series & pie charts (Plotly), STL decomposition, forecasting, outlier detection (LOF), Folium map of price areas (NO1–NO5), correlation heatmaps, snow drift calculations
- **Data sources:** Elhub API, Open-Meteo (ERA5)
- **Tech stack:** Streamlit, Plotly, Pandas, scikit-learn, statsmodels, Folium, River
- **GitHub:** [sigvardbratlie/ind320-dashboard](#)

Rumex Detection Model

YOLO11-based object detection of dock weed (Rumex) in drone imagery for automated grassland monitoring.

- **Key result:** Data augmentation boosted recall from ~5% to ~70% (mAP@50 up to 0.587)
- **Models compared:** YOLO11n vs YOLO11s, with/without heavy augmentation
- **Tech stack:** Ultralytics YOLO11, Albumentations, Label Studio, Jupyter
- **GitHub:** [sigvardbratlie/rumex-detection](#)

Oil Spill Simulator

Python-based mesh-grid simulation of oil spill spread over time, configurable via TOML files for testing detection and thickness estimation algorithms.

- **Key features:** Time-stepped diffusion model, numerical stability checks, batch/single-run support, auto-generated MP4/PNG visualizations
- **Outputs:** Animated videos, image sequences, logs – all saved in dated folders
- **Tech stack:** Python 3.11–3.12, Matplotlib, Opencv, Meshio
- **GitHub:** [sigvardbratlie/oil-spill-sim](#)

Market Data ETL Pipeline

End-to-end scraping & enrichment pipeline for Norwegian marketplace data (Finn.no).

- **Categories:** Cars, homes, rentals, jobs, boats, motorcycles
- **Enrichment:** Geonorge (Nominatim), Statens Vegvesen, Kartverket/Grunnbok
- **Tech stack:** Scrapy + Playwright, XGBoost/CatBoost/LightGBM, Google Cloud (BigQuery, Vertex AI, Kubeflow Pipelines), uv
- **GitHub:** [sigvardbratlie/sibr-market-data-etl](#)

SIBR Market – AI Valuation Platform

Conversational AI platform for real estate and vehicle valuations using LLM agents + public Norwegian registries.

- **Features:** Chat-based valuation, Google OAuth, streaming responses, PDF reports, BigQuery analytics, multi-LLM support (Gemini / GPT-4o / Claude)
 - **Tech stack:** LangGraph agents, Streamlit UI, FastAPI backend, Google Cloud Run, Firestore, WeasyPrint, Plotly
 - **GitHub:** [sigvardbratlie/sibr-market](https://github.com/sigvardbratlie/sibr-market)
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